

Mohammad Mihrab Uddin Chowdhury, Ph.D.

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Summary

- Data scientist with 5 plus years of experience transforming raw data into actionable insights using mechanistic modeling, machine learning, statistical analysis, and data analytics
- Strong leadership and time-management skills evidenced by timely completion of 6 diverse cross-functional collaborative projects and organization of mini-symposia and workshops for professionals across 6 international conferences
- Effective communicator (written and oral) with experience in publishing 6 refereed journal papers and 16 presentations

Technical Skills

Programming Languages: Python, R, SQL, MATLAB, AMPL

Solvers & Libraries: Scikit-Learn, TensorFlow, Pandas, NumPy, Matplotlib, ggplot2, Hugging Face, BERT

Software: SAS, JMP, Tableau, Power BI, Microsoft Excel, Visual Studio Code

Cloud & HPC: AWS (ML Specialty, Solutions Architect, Cloud Practitioner), TTU HPC (SLURM grid search)

Areas of Expertise

Root Cause Analysis, Optimization, Mathematical Programming, Algorithm Design, Simulation, Statistical Analysis, Machine Learning, Deep Learning, Predictive Modeling, Model Validation & Implementation, Big Data, Data Mining, Data Visualization, Data Wrangling, Logistics Optimization, Scheduling, Forecasting, Risk Analysis

Certificates

- Google Data Analytics Professional
- AWS Solutions Architect(Associate)
- AWS ML Specialty
- AWS Cloud Practitioner

Education

PhD & MS Applied Mathematics, Texas Tech University	2024/2021
M.S. & B.S., Mathematics, University of Dhaka, Bangladesh	2018/2016

Courses

Operations Management, Quantitative Financial Management, Principles of Optimization, Statistical Multivariate Analysis, Generative AI with LLM, Intro to Machine Learning for ID Modeling, Statistics and Modeling with Novel Data Streams, Spatial Statistics in Epidemiology and Public Health, The Complete SQL Boot Camp 2022

Selected Project Experience:

Wastewater-Based COVID-19 Forecasting (ML Model with early hospitalization Forecasting)

- Built predictive ML models (Poisson regression, Random Forest) to forecast hospitalizations across 43 ZIP codes and 6 wastewater treatment plant regions in South Carolina
- Integrated multi-source data (wastewater RNA, cases, demographics) into a scalable modeling pipeline
- Engineered lag-based features and identified optimal lag **14-day, improving forecast accuracy to 90%**

Diabetes Risk Prediction (440K+ Records, Imbalanced Data, ML and Augmentation)

- Built classification ML models (Logistic Regression, Gradient Boosting, AdaBoost and Random Forest) on highly imbalanced dataset (86:14 ratio). Applied resampling techniques (ENN, SMOTE variants) **improving recall by 14% from baseline with hyperparameter tuning**
- Identified key risk factors (age, BMI, and blood pressure) aligned with findings

Dengue & Infectious Disease Modeling

- Built data-driven mechanistic models incorporating temperature and spatial factors for forecasting and risk analysis of disease spread across dengue, wildlife diseases, and COVID-19
- Combined mechanistic and statistical approaches for transmission dynamics.
- Generated insights for risk assessment and resource allocation

NYC Restaurant Promotion and Delivery Analytics

- Analyzed **17K+ orders; found 38.8% unrated**: proposed feedback nudges
- Identified weekend demand doubling with **~6-min delays in delivery**: optimized staffing during weekends
- Suggested targeted promotions to mid-tier cuisines and highlight affordable options to maximize revenue
- **Recommended dynamic weekend pricing for high-demand cuisines to improve margins**

ENews Express — A/B Testing for Landing Page Optimization

- Evaluated new vs old landing page performance using statistical hypothesis testing. **In new landing page, conversion rate increased from 40% to 68% ($p < 0.01$) and user engagement increased 37%**
- Recommended deploying the new page to maximize revenue

Professional Experiences

Virginia Tech (VT)
Visiting Assistant Professor

September 2025 - Present

- Developing mechanistic spatial-temporal models using real-world datasets for forecasting and risk analysis of dengue
- Applying machine learning and statistical modeling on 20K+ records to uncover youth suicide risk factors in the U.S.
- Teaching undergraduate courses and managed classrooms of 40+ students

Clemson University
Post-Doctoral Fellow (Research)

June 2024 – September 2025

- Built machine learning models for healthcare forecasting using wastewater and clinical data
- Designed optimization strategies for resource allocation across multiple regions
- Reduced projected hospitalizations through data-driven intervention modeling

Texas Tech University
Graduate Researcher

August 2020 – May 2024

- Developed machine learning models for large-scale healthcare datasets (440K+ records)
- Built mechanistic and simulation models for infectious disease dynamics
- Performed large-scale model training using HPC systems
- Taught undergraduate courses and managed classrooms of 110+ students

Leadership and Managerial Roles

Organizer, Math Career Connection Lecture Series (Grad and UnderGrad), Department of Mathematics, Virginia Tech

January 2026-Present

Judge, iGrads, Graduate Research and discovery symposium, Clemson University

2025

Co- organizer, Session Chair of Mini-symposia and Workshop for professionals in national and international conferences

2022-2026

Selected Honors and Scholarships

Virginia Tech Startup Grant (\$3,000)

Dr. Shelby Hildebrand Fellowship 2023(\$10,000)

GARD2026 Grant Recipient, (\$150)

Create Possible Scholarship - 2021 (\$750)

Halloran Scholar, SISMID 2024 (\$2485)

Hildebrand Scholar - 2021 (\$1500)

Selected Publications

- Chowdhury M.M. et el., Granular insights: A wastewater-based **machine learning** approach for localized COVID-19 hospitalization forecasting, Epidemics Special Issue ‘**AI for ID modeling**’, 2026, Elsevier
- Chowdhury M.M. et el., An Investigation of **Machine Learning** Algorithms and Data Augmentation Techniques for Diabetes Diagnosis Using **Class Imbalanced** BRFSS Dataset, **Healthcare Analytics**, 2024, Elsevier